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Code	<pre> Console Terminal Background Jobs R 4.6.1 ~/Documents/R Programming/NPTEL/ > # --- Lecture 43: Data Frames - Some More Operations --- > # Included here as it is present in the Week 11 notes as well. > > library(MASS) > painters Composition Drawing Colour Expression School Da Udine 10 8 16 3 A Da Vinci 15 16 4 14 A Del Piombo 8 13 16 7 A Del Sarto 12 16 9 8 A Fr. Penni 0 15 8 0 A Gulio Romano 15 16 4 14 A Michelangelo 8 17 4 8 A Perino del Vaga 15 16 7 6 A Perugino 4 12 10 4 A Raphael 17 18 12 18 A F. Zuccaro 10 13 8 8 B Fr. Solviate 13 15 8 8 B Parmigiano 10 15 6 6 B Primaticcio 15 14 7 10 B T. Zucarro 13 14 10 9 B Volterra 12 15 5 8 B Barocci 14 15 6 10 C Cortona 16 14 12 6 C Jusepin 10 10 6 2 C L. Jordaens 13 12 9 6 C Testa 11 15 0 6 C Vanus 15 15 12 13 C Bassano 6 8 17 0 D Bellini 4 6 14 0 D Giorgione 8 9 18 4 D Murillo 6 8 15 4 D Palma Giovane 12 9 14 6 D Palma Vecchio 5 6 16 0 D Pordenone 8 14 17 5 D Tintoretto 15 14 16 4 D Titian 12 15 18 6 D Veronese 15 10 16 3 D Albani 14 14 10 6 E Caravaggio 6 6 16 0 E Corregio 13 13 15 12 E Domenichino 15 17 9 17 E Guercino 18 10 10 4 E Lanfranco 14 13 10 5 E The Carracci 15 17 13 13 E Durer 8 10 10 8 F </pre>
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Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/
Holbein      9      10      16      13      F
Pourbus      4      15      6       6      F
Van Leyden   8       6       6       4      F
Diepenbeck  11      10      14       6      G
J. Jordaens  10       8      16       6      G
Otho Venius  13      14      10      10      G
Rembrandt   15       6      17      12      G
Rubens      18      13      17      17      G
Teniers     15      12      13       6      G
Van Dyck    15      10      17      13      G
Bourdon     10       8       8       4      H
Le Brun     16      16      8      16      H
Le Suer     15      15      4      15      H
Poussin     15      17      6      15      H
> summary painters$School
  A  B  C  D  E  F  G  H
10  6  6 10  7  4  7  4
> attach(painters)
> summary(School)
  A  B  C  D  E  F  G  H
10  6  6 10  7  4  7  4
> summary(Composition)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.00   8.25   12.50   11.56   15.00   18.00
> detach(painters)
>
> subset(painters, School=="F")
      Composition Drawing Colour Expression School
Durer           8      10      10           8      F
Holbein          9      10      16          13      F
Pourbus          4      15       6           6      F
Van Leyden       8       6       6           4      F
> painters[painters[["School"]] == "F", ]
      Composition Drawing Colour Expression School
Durer           8      10      10           8      F
Holbein          9      10      16          13      F
Pourbus          4      15       6           6      F
Van Leyden       8       6       6           4      F
> subset(painters, Composition <= 6)
      Composition Drawing Colour Expression School
Fr. Penni        0      15       8           0      A
Perugino         4      12      10           4      A
Bassano          6       8      17           0      D
Bellini          4       6      14           0      D
Murillo          6       8      15           4      D
Palma Vecchio    5       6      16           0      D
Caravaggio       6       6      16           0      E
Pourbus          4      15       6           6      F
> subset(painters, School=="F", select=c(-3,-5))
      Composition Drawing Expression
Durer           8      10           8
Holbein          9      10          13
Pourbus          4      15           6
Van Leyden       8       6           4
> splitted = split(painters, painters$School)
> splitted
$A
      Composition Drawing Colour Expression School
Da Udine        10       8      16           3      A
Da Vinci        15      16       4          14      A
Del Piombo       8      13      16           7      A
Del Sarto       12      16       9           8      A
Fr. Penni        0      15       8           0      A
Giulio Romano   15      16       4          14      A
Michelangelo     8      17       4           8      A
Perino del Vaga  15      16       7           6      A
Perugino         4      12      10           4      A
Raphael        17      18      12          18      A
$B
      Composition Drawing Colour Expression School
F. Zucarro       10      13       8           8      B
Fr. Salviata     13      15       8           8      B
Parmigiano      10      15       6           6      B
Primaticcio     15      14       7          10      B
T. Zucarro       13      14      10           9      B
Volterra        12      15       5           8      B

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Console | Terminal | Background Jobs
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$C
      Composition Drawing Colour Expression School
Barocci      14      15      6      10      C
Cortona      16      14      12      6      C
Josepin      10      10      6      2      C
L. Jordaens  13      12      9      6      C
Testa        11      15      0      6      C
Vanius       15      15      12      13      C

$D
      Composition Drawing Colour Expression School
Bassano      6      8      17      0      D
Bellini      4      6      14      0      D
Giorgione    8      9      18      4      D
Murillo      6      8      15      4      D
Palma Giovane 12      9      14      6      D
Palma Vecchio 5      6      16      0      D
Pordenone    8      14      17      5      D
Tintoretto   15      14      16      4      D
Titian       12      15      18      6      D
Veronese     15      10      16      3      D

$E
      Composition Drawing Colour Expression School
Albani       14      14      10      6      E
Caravaggio   6      6      16      0      E
Correggio    13      13      15      12      E
Domenichino  15      17      9      17      E
Guercino     18      10      10      4      E
Lanfranco    14      13      10      5      E
The Carraci  15      17      13      13      E

$F
      Composition Drawing Colour Expression School
Durer        8      10      10      8      F
Holbein      9      10      16      13      F
Poussin      4      15      6      6      F
Van Leyden   8      6      6      4      F

$G
      Composition Drawing Colour Expression School
Diepenbeck   11      10      14      6      G
J. Jordaens  10      8      16      6      G
Otho Ventius 13      14      10      10      G
Rembrandt    15      6      17      12      G
Rubens       18      13      17      17      G
Teniers      15      12      13      6      G
Van Dyck     15      10      17      13      G

$H
      Composition Drawing Colour Expression School
Bourdon      10      8      8      4      H
Le Brun      16      16      8      16      H
Le Suer      15      15      4      15      H
Poussin      15      17      6      15      H

> is.data.frame(splitted$A)
[1] TRUE
> # --- Lecture 46: Importing and Reading Excel and Other Data Files ---
>
> # Set working directory as shown in the notes
> # setwd("C:/RCourse/")
>
> # Install once if required:
> # install.packages("readxl")
> library(readxl)
>
> read_excel("datafile.xlsx")
Error: 'path' does not exist: 'datafile.xlsx'
Show Traceback
Rerun with Debug

> # Example Excel file from the notes
> dataspxcel = read_excel("spexcel.xlsx", sheet=1)
Error: 'path' does not exist: 'spexcel.xlsx'
Show Traceback
Rerun with Debug

> # Read a limited number of rows
> read_excel(datasets, n_max=3)
Error: object 'datasets' not found
Show Traceback
Rerun with Debug

> # HTML table data
> # install.packages("XML")
> library(XML)
Error in library(XML) : there is no package called 'XML'
Show Traceback
Rerun with Debug

> |
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Environment			History	Connections	Git	Tutorial
Import Dataset			28 MiB			
R			Global Environment			
Data						
ab	List of 6					
datemat	chr [1:4, 1:3]	"2020" "2021" "2022" "2023" "07" "08" "09" "10" "2...				
datematrix	num [1:4, 1:3]	2020 2021 2022 2023 7 ...				
datesplt	List of 4					
df1	4 obs. of 2 variables					
df11	4 obs. of 2 variables					
df2	4 obs. of 3 variables					
df22	3 obs. of 2 variables					
list1	List of 6					
list12	List of 6					
list2	List of 6					
splitted	List of 8					
Values						
brands	chr [1:7]	"A" "A" "B" "B" "B" "B" "C"				
brands_fac	Factor w/ 3 levels	"A","B","C": 1 1 2 2 2 2 3				
colours	chr [1:3]	"blue" "green" "red"				
d		"Thu Oct 1 14:08:05 2026"				
dates	chr [1:4]	"2020-07-24" "2021-08-25" "2022-09-26" "2023-10-27"				
evenno	num [1:5]	2 4 6 8 10				
facy	Factor w/ 6 levels	"one","two","three",...: 1 4 3 5 4 2 4				
income	Ord.factor w/ 3 levels	"low"<"medium"<...: 3 3 1 2 2				
labels.dieface	chr [1:6]	"one" "two" "three" "four" "five" "six"				
names	chr [1:3]	"Prof. Singh" "Mr. Venkat" "Dr. Jha"				
possible.dieface	num [1:6]	1 2 3 4 5 6				
str	chr [1:3]	"R Course" "exercises" "include examples of R language"				
x		"R course 24.07.2021"				
y		"Number of participants: 25"				
z		"INDIAN INSTITUTE OF TECHNOLOGY"				
z1	num [1:3]	1.1 2.22 3.33				